

Biology, 2018 EC, entrance exam, version 2

Grade 12 · Biology · 2018 E.C.

Time allowed: 120 minutes

Questions: 98

Total marks: 98

Name _____ Grade _____ ID No. _____ Date _____

1. Which of the following is a major carbohydrate found in plants that humans cannot digest? (1)
 - A. Glycogen
 - B. Fructose
 - C. Cellulose
 - D. Maltose
2. Which of the following best describes the difference between saturated and unsaturated fatty acids? (1)
 - A. Saturated fatty acids have one or more double bonds between carbon atoms, while unsaturated fatty acids have only single bonds.
 - B. Saturated fatty acids are typically found in plants, while unsaturated fatty acids are found in animals.
 - C. Saturated fatty acids have no double bonds between carbon atoms, while unsaturated fatty acids have one or more double bonds.
 - D. Saturated fatty acids are liquid at room temperature, while unsaturated fatty acids are solid.
3. What is the role of bile in digestion? (1)
 - A. It breaks down proteins in the stomach.
 - B. It emulsifies fats, breaking them down into smaller droplets.
 - C. It helps to digest carbohydrates in the small intestine.
 - D. It neutralizes the acid from the stomach to make the food alkaline.
4. Which part of the digestive system is responsible for producing enzymes that digest carbohydrates, proteins, and fats? (1)
 - A. Stomach
 - B. Small intestine
 - C. Pancreas
 - D. Liver
5. Which structure separates the thoracic cavity from the abdominal cavity and plays a key role in breathing? (1)
 - A. Larynx
 - B. Diaphragm
 - C. Trachea
 - D. Bronchi
6. What is the main product of anaerobic respiration in muscle cells during vigorous exercise? (1)
 - A. Lactic acid
 - B. Ethanol
 - C. Oxygen
 - D. Carbon dioxide
7. Which of the following best describes the process of fermentation in yeast cells? (1)
 - A. Yeast cells use oxygen to break down glucose and produce ATP, water, and carbon dioxide.
 - B. Yeast cells break down glucose without oxygen, producing ethanol and carbon dioxide.
 - C. Yeast cells produce ATP by converting lactic acid into energy.
 - D. Yeast cells use lactic acid as the main energy source in the absence of oxygen.

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8. Which of the following statements about red blood cells is true? (1)
- A. Red blood cells have a nucleus and live for about 10-20 days.
 - B. Red blood cells are shaped as biconcave discs, which gives them a large surface area for oxygen exchange.
 - C. Red blood cells are responsible for fighting infections in the body.
 - D. Red blood cells are made in the liver and transport carbon dioxide around the body.
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9. In the ABO blood group system, which blood type can receive blood from any of the four ABO blood types? (1)
- A. Type A
 - B. Type B
 - C. Type AB
 - D. Type O
-
10. Which of the following is NOT a characteristic of bacteria? (1)
- A. Bacteria are single-celled organisms.
 - B. Bacterial cells contain a nucleus to store genetic material.
 - C. Bacteria can have structures like flagella to help them move.
 - D. Some bacteria are useful for processes like fermentation and decay.
-
11. What is the main purpose of using UHT (Ultra High Temperature) treatment for food? (1)
- A. To make the food taste better
 - B. To keep the food warm for longer
 - C. To sterilize the food by killing all micro-organisms
 - D. To preserve the food by freezing
-
12. Which of the following is the most common way people get infected with tapeworms? (1)
- A. By drinking contaminated water
 - B. By eating undercooked or raw meat containing young tapeworms
 - C. By coming into contact with contaminated soil
 - D. Through insect bites
-
13. Which of the following is the most effective way to prevent the spread of gonorrhea? (1)
- A. Getting vaccinated for gonorrhea
 - B. Using a male or female condom during sexual activity
 - C. Avoiding alcohol consumption
 - D. Avoiding all contact with infected individuals
-
14. What is the primary reason why chancroid is particularly concerning in regions with high HIV/AIDS rates? (1)
- A. It causes severe pain and tissue damage
 - B. It can lead to permanent infertility
 - C. The sores caused by chancroid increase the likelihood of contracting HIV
 - D. It has no effective treatment
-
15. What is the primary role of neurotransmitters at a synapse? (1)
- A. To produce electrical impulses
 - B. To cross the synapse and start an electrical impulse in the next neuron
 - C. To insulate the axon for faster impulse transmission
 - D. To repair damaged neurons

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- 16.** What is a conditioned reflex action? (1)
- A. A reflex action that is innate and present at birth
 - B. A reflex action that is learned through experience
 - C. A reflex action that occurs only in response to physical injury
 - D. A reflex action that is triggered by emotional stimuli
-
- 17.** Which of the following conditions can be corrected using concave (diverging) lenses? (1)
- A. Long sight
 - B. Short sight
 - C. Astigmatism
 - D. Presbyopia
-
- 18.** Which of the following is a function of the epidermis layer of the skin? (1)
- A. Temperature regulation
 - B. Protection against UV light
 - C. Containing blood vessels and sweat glands
 - D. Storing fat for insulation
-
- 19.** Which of the following is a primary function of insulin in the body? (1)
- A. To increase blood sugar levels
 - B. To regulate the heart rate
 - C. To lower blood sugar levels by helping glucose enter cells
 - D. To speed up the breathing rate
-
- 20.** Which of the following contraceptive methods is considered to be 99% effective at preventing pregnancy and does not require daily action from the user? (1)
- A. The combined oral contraceptive pill
 - B. Male condoms
 - C. Hormonal implants
 - D. Diaphragm
-
- 21.** What is the primary function of sweating in homoiotherms? (1)
- A. To increase body temperature
 - B. To release excess carbon dioxide
 - C. To cool the body down by evaporation
 - D. To conserve water and ions
-
- 22.** Which physiological response helps conserve heat when the core body temperature drops? (1)
- A. Sweating
 - B. Vasodilation
 - C. Vasoconstriction
 - D. Panting
-
- 23.** Which of the following structures in a leaf is primarily responsible for controlling gas exchange and water loss? (1)
- A. Palisade mesophyll
 - B. Stomata
 - C. Xylem
 - D. Phloem
-
- 24.** Which of the following is NOT a requirement for photosynthesis? (1)
- A. Water
 - B. Carbon dioxide
 - C. Chlorophyll
 - D. Nitrogen
-
- 25.** What is the main purpose of boiling a leaf in water during the starch test for photosynthesis? (1)
- A. To remove the starch
 - B. To soften the leaf
 - C. To kill the leaf and stop biochemical processes
 - D. To add iodine solution to the leaf

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- 34.** Which of the following is a characteristic of an enzyme's active site? (1)
- A. It is where the enzyme binds to its cofactor.
 - B. It is the region where the enzyme binds to its substrate.
 - C. It binds to non-protein molecules like vitamins.
 - D. It is only responsible for enzyme activation.
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- 35.** What is the primary function of oxidoreductase enzymes? (1)
- A. Catalyze the transfer of functional groups.
 - B. Catalyze the breakdown of starch.
 - C. Catalyze oxidation-reduction reactions by transferring electrons.
 - D. Catalyze the joining of two molecules using ATP.
-
- 36.** Which factor does NOT affect enzyme activity? (1)
- A. Temperature
 - B. pH
 - C. Enzyme concentration
 - D. The color of the enzyme
-
- 37.** What happens when an enzyme is exposed to extreme pH or temperature? (1)
- A. The enzyme becomes more efficient.
 - B. The enzyme becomes denatured and loses its function.
 - C. The enzyme speeds up the reaction rate.
 - D. The enzyme increases its affinity for the substrate.
-
- 38.** Which of the following is an example of a cofactor required by some enzymes for activity? (1)
- A. Substrate
 - B. Vitamin B complex (Coenzyme)
 - C. Product
 - D. Inhibitor
-
- 39.** What is the main difference between the structure of DNA and RNA? (1)
- A. DNA has a single strand, while RNA has a double strand
 - B. DNA contains uracil, while RNA contains thymine
 - C. DNA contains deoxyribose, while RNA contains ribose
 - D. DNA contains ribose, while RNA contains deoxyribose
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- 40.** What is the function of RNA in cells? (1)
- A. RNA is responsible for storing genetic information
 - B. RNA facilitates the translation of DNA into proteins
 - C. RNA stores energy for the cell
 - D. RNA is responsible for DNA replication
-
- 41.** What is the primary function of meiosis in organisms? (1)
- A. To produce genetically identical daughter cells for growth and repair
 - B. To produce gametes (sex cells) with half the number of chromosomes
 - C. To replicate DNA for cellular repair
 - D. To divide somatic cells for tissue regeneration

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- 42.** What happens during the S phase of the cell cycle? (1)
- A. The cell grows and accumulates nutrients
 - B. DNA is replicated to form sister chromatids
 - C. The nuclear envelope dissolves, and chromosomes condense
 - D. The cytoplasm divides into two separate cells
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- 43.** In a monohybrid cross, when two heterozygous individuals (Tt x Tt) are crossed, what is the expected phenotypic ratio of the offspring? (1)
- A. 1:1
 - B. 3:1
 - C. 2:1
 - D. 1:2:1
-
- 44.** Which of the following statements best describes Mendel's Law of Independent Assortment? (1)
- A. Alleles for a single trait separate during meiosis
 - B. Alleles for different traits separate independently of each other during meiosis
 - C. Dominant traits will always be expressed in offspring
 - D. Recessive traits will always be expressed when two recessive alleles are inherited
-
- 45.** What is the genetic composition that determines an individual to be male in humans? (1)
- A. XX
 - B. XY
 - C. X
 - D. YY
-
- 46.** Which of the following is an example of non-Mendelian inheritance? (1)
- A. Complete dominance
 - B. Incomplete dominance
 - C. Gene segregation
 - D. Independent assortment
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- 47.** What is one of the main uses of biotechnology in food production? (1)
- A. Genetic engineering of crops
 - B. Creating synthetic chemicals
 - C. Developing biological weapons
 - D. Reducing biodiversity loss
-
- 48.** Which of the following is a key feature of forensic biotechnology? (1)
- A. The use of genetic engineering to improve crops
 - B. Analysis of DNA and other biological evidence at crime scenes
 - C. The development of new biofuels
 - D. Engineering microorganisms for industrial use
-
- 49.** What is the main role of biotechnology in medicine? (1)
- A. Increasing the cost of drug production
 - B. Providing effective diagnostics, prevention, and treatment
 - C. Reducing the need for medical research
 - D. Limiting the development of new disease
-
- 50.** Which of the following is an example of a biotechnological process used for wastewater treatment? (1)
- A. PCR amplification
 - B. Bio-trickling filters
 - C. Genetic engineering of plants
 - D. DNA sequencing

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- 51.** According to the World Health Organization (WHO), health is defined as: (1)
- A. The absence of disease
 - B. A state of complete physical, mental and social well-being
 - C. The ability to grow and reproduce
 - D. The ability to exercise regularly
-
- 52.** What is the primary role of biology in nature conservation? (1)
- A. To reduce pollution
 - B. To manage renewable and non-renewable resources properly
 - C. To increase the demand for natural resources
 - D. To exploit natural resources for immediate use
-
- 53.** Which of the following is an example of a non-renewable resource? (1)
- | | |
|-----------|----------|
| A. Cattle | B. Trees |
| C. Coal | D. Water |
-
- 54.** Which of the following activities does not contribute to the appreciation of nature? (1)
- | | |
|--------------------|---|
| A. Planting a tree | B. Reducing pollution |
| C. Hugging a tree | D. Urbanization and excessive development |
-
- 55.** What does food security ensure? (1)
- A. That food is produced only for wealthy individuals
 - B. That everyone has physical, social, and economic access to nutritious food
 - C. That food prices are high to prevent overconsumption
 - D. That food production remains the same each year
-
- 56.** Which of the following is a renewable resource in Ethiopia? (1)
- | | |
|-----------|---------|
| A. Gold | B. Coal |
| C. Coffee | D. Oil |
-
- 57.** Which of the following is NOT a main group of microorganisms? (1)
- | | |
|-------------|------------|
| A. Bacteria | B. Fungi |
| C. Mammals | D. Viruses |
-
- 58.** What is the key difference between prokaryotic and eukaryotic cells? (1)
- A. Eukaryotic cells have no nucleus, while prokaryotic cells have a nucleus.
 - B. Eukaryotic cells are unicellular, while prokaryotic cells are multicellular.
 - C. Prokaryotic cells have no membrane-enclosed nucleus, while eukaryotic cells do.
 - D. Eukaryotic cells lack cell membranes, while prokaryotic cells have them.
-
- 59.** Which shape do cocci bacteria have? (1)
- | | |
|------------------|----------------|
| A. Spiral-shaped | B. Rod-shaped |
| C. Spherical | D. Oval-shaped |

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- 60.** Which of the following best describes the habitat of Archaea? (1)
- A. Archaea are only found in warm, tropical climates.
 - B. Archaea inhabit extreme environments such as hot springs, saline lakes, and deep-sea vents.
 - C. Archaea only live in the human gut.
 - D. Archaea thrive in forested areas with abundant rainfall.
-
- 61.** What is the primary difference between fungi and plants? (1)
- A. Fungi can photosynthesize, while plants cannot.
 - B. Fungi do not have cell walls, while plants do.
 - C. Fungi have chitin in their cell walls, while plants have cellulose.
 - D. Fungi are always unicellular, while plants are always multicellular.
-
- 62.** Which of the following is an example of how fungi benefit humans? (1)
- A. Fungi can cause diseases in humans.
 - B. Fungi help in the fermentation process to produce traditional beverages like Tej and Tella.
 - C. Fungi only serve as food for other organisms.
 - D. Fungi are used to produce energy in power plants.
-
- 63.** Which of the following diseases is caused by the protozoan Plasmodium? (1)
- | | |
|------------------|---------------|
| A. Giardiasis | B. Malaria |
| C. Toxoplasmosis | D. Amoebiasis |
-
- 64.** What is the primary mode of transmission for Leishmania parasites? (1)
- | | |
|-----------------------|----------------------------------|
| A. Contaminated water | B. Mosquito bites |
| C. Sand fly bites | D. Direct human-to-human contact |
-
- 65.** Which of the following protozoan diseases affects the liver and causes diarrhea in humans? (1)
- | | |
|---------------|----------------------------|
| A. Malaria | B. Toxoplasmosis |
| C. Amoebiasis | D. African trypanosomiasis |
-
- 66.** What is the outer protein shell of a virus called? (1)
- | | |
|-------------|-------------|
| A. Envelope | B. Genome |
| C. Capsid | D. Peplomer |
-
- 67.** Which of the following types of viruses contain RNA as their genetic material? (1)
- | | |
|-----------------|----------------|
| A. Papovavirus | B. Parvovirus |
| C. Retroviruses | D. Herpesvirus |
-
- 68.** In which of the following structures do DNA viruses primarily replicate? (1)
- | | |
|--------------|-----------------|
| A. Cytoplasm | B. Nucleus |
| C. Ribosomes | D. Mitochondria |
-
- 69.** Which of the following is NOT a stage in the viral replication cycle? (1)
- A. Attachment of the virus to a host cell
 - B. Replication of the viral genome
 - C. Fusion of the viral particle with a host's DNA
 - D. Release of new virus particles

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- 70.** What role do microorganisms play in agriculture? (1)
- A. They help in the decomposition of organic matter and formation of humus
 - B. They prevent the nitrogen cycle from occurring
 - C. They convert phosphorus to a form that plants cannot use
 - D. They eliminate all plant pests
-
- 71.** What is the main benefit of using anaerobic bacteria in sewage treatment? (1)
- A. They produce oxygen for the process
 - B. They reduce the volume of sludge and produce methane gas, which can be used as an energy source
 - C. They increase the oxygen consumption of the water
 - D. They prevent bacteria from spoiling the water
-
- 72.** Which of the following is an example of disease transmission via droplet infection? (1)
- | | |
|--------------|---------------|
| A. Cholera | B. Malaria |
| C. Pneumonia | D. Salmonella |
-
- 73.** Which of the following is an example of a disaccharide? (1)
- | | |
|------------|--------------|
| A. Glucose | B. Maltose |
| C. Starch | D. Cellulose |
-
- 74.** What is the main difference between alpha-D-glucose and beta-D-glucose? (1)
- A. The number of carbon atoms
 - B. The position of the hydroxyl group on carbon 1
 - C. The type of glycosidic bond formed
 - D. The size of the glucose molecule
-
- 75.** Which of the following is a polysaccharide that is a storage form of glucose in animals? (1)
- | | |
|--------------|-------------|
| A. Starch | B. Glycogen |
| C. Cellulose | D. Chitin |
-
- 76.** What type of reaction results in the formation of disaccharides from monosaccharides? (1)
- | | |
|---------------|-----------------|
| A. Hydrolysis | B. Condensation |
| C. Oxidation | D. Reduction |
-
- 77.** Which of the following lipids are made up of fatty acids, glycerol or other alcohol, and phosphoric acid? (1)
- | | |
|------------------|-----------|
| A. Triglycerides | B. Waxes |
| C. Phospholipids | D. Sterol |
-
- 78.** What is the main reason lipids are insoluble in water? (1)
- A. They contain a large number of oxygen atoms.
 - B. They are polar molecules.
 - C. They are hydrophobic in nature.
 - D. They have a high molecular weight.
-
- 79.** Which type of lipid is an ester of fatty acids and long-chain alcohols? (1)
- | | |
|------------------|-------------|
| A. Triglycerides | B. Waxes |
| C. Phospholipids | D. Steroids |

-
- 80.** What is the primary function of glycolipids in the cell membrane? (1)
- A. To store energy
 - B. To provide insulation
 - C. To maintain the stability of the membrane and facilitate cellular recognition
 - D. To act as an energy source
-
- 81.** What is the primary function of the Calvin cycle in photosynthesis? (1)
- A. To split water molecules
 - B. To convert light energy into chemical energy
 - C. To fix carbon and produce glucose
 - D. To generate oxygen gas
-
- 82.** Which pigment is most responsible for the absorption of light energy in photosynthesis? (1)
- | | |
|------------------|------------------|
| A. Carotenoids | B. Phycobilins |
| C. Chlorophyll a | D. Chlorophyll b |
-
- 83.** Where does the light-dependent reaction of photosynthesis take place? (1)
- A. In the stroma of the chloroplast
 - B. In the cytoplasm
 - C. In the thylakoid membranes of the chloroplast
 - D. In the mitochondria
-
- 84.** What is the role of NADPH in the light-dependent reactions of photosynthesis? (1)
- A. To produce ATP
 - B. To provide reducing power for the Calvin cycle
 - C. To absorb light energy
 - D. To split water molecules
-
- 85.** Which of the following occurs during the light-independent reactions (Calvin cycle)? (1)
- A. Water is split to release oxygen
 - B. Light energy is converted into chemical energy
 - C. Carbon dioxide is fixed into an organic molecule
 - D. ATP is converted into NADP+
-
- 86.** Which of the following plants uses the C4 pathway for carbon fixation? (1)
- | | |
|--------------|-------------|
| A. Peanuts | B. Cacti |
| C. Sugarcane | D. Soybeans |
-
- 87.** What is the primary reason that C4 and CAM plants are more efficient in hot, dry climates? (1)
- A. They have larger stomata
 - B. They reduce photorespiration by increasing CO₂ concentration in their cells
 - C. They do not require light for photosynthesis
 - D. They only perform photosynthesis at night

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- 88.** What does photorespiration involve in C3 plants? (1)
- A. The enzyme rubisco adds O₂ instead of CO₂ to RuBP, wasting energy
 - B. The plant fixes CO₂ twice in one cycle
 - C. The plant produces excess sugars that are stored for later use
 - D. The plant converts light energy directly into ATP without CO₂ fixation
-
- 89.** Which of the following is a direct consequence of photosynthesis for life on Earth? (1)
- A. It contributes to the oxygen balance in the atmosphere
 - B. It releases carbon dioxide into the atmosphere
 - C. It consumes all atmospheric carbon
 - D. It destroys fossil fuels
-
- 90.** Where does glycolysis occur in a eukaryotic cell? (1)
- | | |
|-----------------|--------------|
| A. Mitochondria | B. Cytosol |
| C. Nucleus | D. Ribosomes |
-
- 91.** What is the main role of ATP in cellular metabolism? (1)
- A. To store energy in chemical bonds
 - B. To transport oxygen in the cell
 - C. To store and transfer energy for biochemical reactions
 - D. To break down glucose
-
- 92.** Which of the following is a product of the Krebs cycle? (1)
- | | |
|-------------|---|
| A. Oxygen | B. NADH, FADH ₂ , ATP, and CO ₂ |
| C. Pyruvate | D. Glucose |
-
- 93.** What occurs during the pyruvate oxidation (link reaction)? (1)
- A. Glucose is broken into two pyruvate molecules
 - B. Pyruvate is converted into acetyl CoA
 - C. Oxygen is consumed to form water
 - D. ATP is synthesized directly
-
- 94.** What is the final electron acceptor in the electron transport chain (ETC)? (1)
- | | |
|----------------------|------------|
| A. NADH | B. Oxygen |
| C. FADH ₂ | D. Glucose |
-
- 95.** Which part of the mitochondria is involved in oxidative phosphorylation? (1)
- | | |
|-------------------|-------------------|
| A. Outer membrane | B. Inner membrane |
| C. Matrix | D. Cristae |
-
- 96.** What is the main purpose of a respirometer? (1)
- A. To measure the volume of carbon dioxide produced during respiration
 - B. To measure the temperature change during aerobic respiration
 - C. To measure the volume of oxygen consumed during aerobic respiration
 - D. To measure the amount of glucose used in the respiration process

97. What happens during fermentation in the absence of oxygen? (1)

- A. Oxygen is consumed to produce ATP
- B. Pyruvate is converted to lactate or ethanol to regenerate NAD⁺
- C. The electron transport chain produces ATP
- D. Glucose is completely oxidized to carbon dioxide and water

98. Which type of fermentation occurs in muscle cells when oxygen is scarce? (1)

- | | |
|-----------------------------|------------------------|
| A. Alcohol fermentation | B. Aerobic respiration |
| C. Lactic acid fermentation | D. Glycolysis |

— END OF EXAMINATION —

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